

This document summarises the compliance obligations that apply to solar PV, wind, and battery-storage plants in each supported country. It is derived from the ShamsIQ compliance packs shipped with the platform. For each jurisdiction we show the technical grid-code envelope the plant must meet (ride-through, frequency, reactive power), the metering and data-retention requirements, the periodic reports you must file with the regulator and grid operator, and the sources behind every rule so you can trace back to the authoritative text.

How to use this guide: onboard a plant with the correct country and ShamsIQ automatically binds the plant to the pack below. The reporting module will then include the applicable obligations in every generated report. This document is your reference for *why* those obligations apply — the clauses and authorities behind each rule.

Full glossary

Each country section below has its own focused key-terms box. This master list covers every term used across the guide.

▼ Show all 39 terms

LVRT — Low Voltage Ride-Through

The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

PCC — Point of Common Coupling

The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

FRT — Fault Ride-Through

Umbrella term for LVRT + HVRT — staying connected through faults. Prevents simultaneous trips of renewable plants during grid disturbances.

MPPT — Maximum Power Point Tracking

Controller in the inverter that tunes the DC operating voltage/current to extract the most power from the PV string given irradiance and temperature.

CEI 0-16

Italian technical standard for MV/HV grid connection. Defines fault ride-through curves, reactive power requirements, and Interface Protection System (SPI).

CEI 0-21

Italian standard for LV grid connection (< 400 V). Simpler envelope than CEI 0-16 — targets residential + small commercial.

P.O. 12.3

Spanish Procedimiento de Operación 12.3 — technical operating procedure issued by REE that defines ride-through and reactive power behaviour at the PCC.

RD 244/2019

Spanish Royal Decree governing self-consumption: registration, metering, compensation mechanism (simplified or surplus sold), and reporting duties.

CECRE — Centro de Control de Energías Renovables

REE control centre for renewables in Spain. Plants ≥ 1 MW must stream real-time telemetry here.

CNMC — Comisión Nacional de los Mercados y la Competencia

Spanish national markets & competition regulator. Oversees electricity market, settlement, and self-consumption declarations.

GSE — Gestore dei Servizi Energetici

Italian state-owned body that administers renewable incentive schemes (SSP, RID) and collects monthly production telemetry.

ARERA — Autorità di Regolazione per Energia Reti e Ambiente

Italian energy, networks, and environment regulator.

Terna

Italian transmission system operator (TSO). Runs the HV grid and publishes grid-code annexes (CEI 0-16).

REE — Red Eléctrica de España

Spanish TSO. Operates the transmission grid and CECRE.

SSP — Scambio Sul Posto

Italian "on-site exchange" net-metering scheme administered by GSE for small self-consumption plants.

RID — Ritiro Dedicato

Italian dedicated withdrawal scheme — GSE purchases excess energy from qualifying plants at regulated prices.

DEWA — Dubai Electricity & Water Authority

Integrated utility for Dubai; runs distribution + Shams Dubai net-metering.

EWEC — Emirates Water & Electricity Company

Abu Dhabi utility offtaker and single buyer.

Shams Dubai

DEWA's distributed solar programme. Net-metered rooftop/commercial PV with technical requirements defined by DEWA.

RSB — Regulatory & Supervisory Bureau

Dubai regulator for electricity, water, and district cooling.

WERA — Water and Electricity Regulatory Authority

Saudi Arabian energy regulator (formerly ECRA).

SEC — Saudi Electricity Company

Saudi utility operating transmission and distribution.

SPPC — Saudi Power Procurement Company

Single-buyer offtaker for utility-scale IPPs in Saudi Arabia.

SSPV — Small-Scale Solar PV

Saudi regulatory framework for distributed rooftop/commercial PV with net billing (not net metering).

EPRA — Energy and Petroleum Regulatory Authority

Kenyan energy sector regulator.

KPLC — Kenya Power & Lighting Company

Kenyan single-buyer utility and offtaker for PPAs.

NEMA — National Environment Management Authority

Kenyan environmental regulator — reviews EIAs and annual compliance audits.

NERC — Nigerian Electricity Regulatory Commission

Federal regulator for the Nigerian electricity industry.

NBET — Nigerian Bulk Electricity Trading

Single-buyer bulk trader that signs PPAs with IPPs in Nigeria.

NEMSA — Nigerian Electricity Management Services Agency

Nigerian electrical safety agency — mandates pre-energisation inspection.

NERSA — National Energy Regulator of South Africa

South African electricity, gas, and piped-fuel regulator.

SSEG — Small-Scale Embedded Generation

South African category for distributed generation ≤ 1 MW connected at LV/MV. Governed by NRS 097-2-1 and NRS 097-2-3.

NRS 097

South African standard series for grid connection of embedded generation. -2-1 covers utility interface, -2-3 covers simplified registration for ≤ 100 kVA.

RPP — Renewable Power Plant

South African grid-code category covering larger renewable plants (> 1 MW). Sub-categorised A1/A2/A3/B/C by capacity with escalating requirements.

REIPPPP — Renewable Energy Independent Power Producer Procurement Programme

South African competitive procurement programme for utility-scale renewables, overseen by DMRE.

COC — Certificate of Compliance

South African post-installation certificate issued by an accredited electrician. Must be revalidated after modifications.

GoO — Guarantees of Origin

EU tradable certificate attesting that 1 MWh of electricity was generated from renewables. Issued via national registries (REN in PT, GSE in IT, CNMC in ES).

DiCo — Dichiarazione di Conformità

Italian declaration of conformity issued by the installer at commissioning. Required for grid connection.

OCA — Organismo de Control Autorizado

Spanish accredited inspection body. Required 5-yearly inspection for installations > 100 kW.

Countries covered

Europe

[Spain \(ES\)](#)

[Portugal \(PT\)](#)

[Italy \(IT\)](#)

Sub-Saharan Africa

[Kenya \(KE\)](#)

[Nigeria \(NG\)](#)

[South Africa \(ZA\)](#)

Middle East

[United Arab Emirates \(AE\)](#)

[Saudi Arabia \(SA\)](#)

Spain (ES)

Pack v2026.Q2 · Currency EUR · Timezone Europe/Madrid

Regulator: [CNMC](#) · Grid operator: [Red Eléctrica de España \(REE\)](#)

Spain separates technical grid operation (REE / CECRE) from market and competition oversight (CNMC). Self-consumption was liberalised by RD 244/2019 and now covers most rooftop and commercial installations under either simplified compensation or surplus-sale regimes. Larger plants (≥ 1 MW) are dispatched through CECRE and must obey the Procedimientos de Operación — notably P.O. 12.3 for fault ride-through.

Key terms for Spain

REE — Red Eléctrica de España

Spanish TSO. Operates the transmission grid and CECRE.

CNMC — Comisión Nacional de los Mercados y la Competencia

Spanish national markets & competition regulator. Oversees electricity market, settlement, and self-consumption declarations.

CECRE — Centro de Control de Energías Renovables

REE control centre for renewables in Spain. Plants ≥ 1 MW must stream real-time telemetry here.

P.O. 12.3

Spanish Procedimiento de Operación 12.3 — technical operating procedure issued by REE that defines ride-through and reactive power behaviour at the PCC.

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The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

PCC — Point of Common Coupling

The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

GoO — Guarantees of Origin

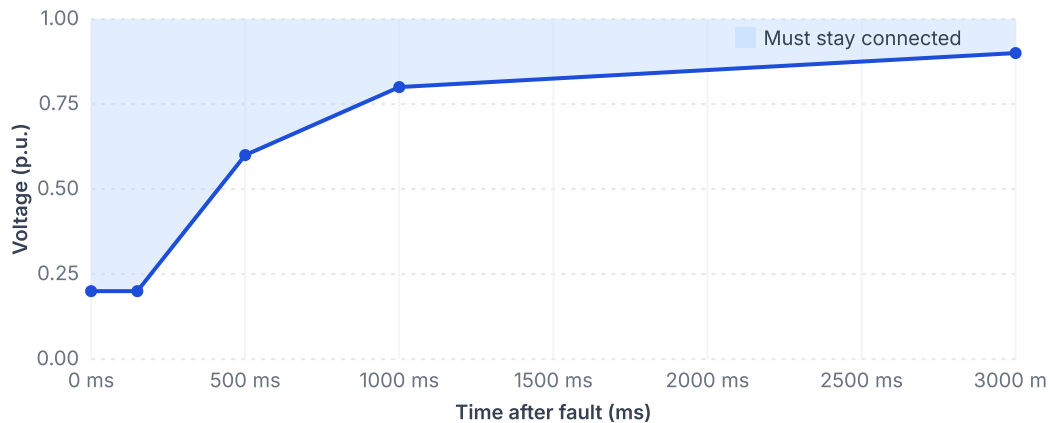
EU tradable certificate attesting that 1 MWh of electricity was generated from renewables. Issued via national registries (REN in PT, GSE in IT, CNMC in ES).

Grid connection code

Reference: P.O. 12.2 / P.O. 12.3 / RD 413/2014 / RD 244/2019

Plants ≥ 1 MW must stream real-time telemetry to CECRE.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

UNE 206007-1 / IEC 62116

RAMP-RATE LIMIT

10% nominal/min

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

10 years

CALIBRATION

every 60 months

Reporting obligations

monthly	Monthly self-consumption de... → Distributor / CNMC · 20d CECRE real-time telemetry s... → REE (CECRE) · 10d
annual	Annual production report → CNMC · 90d

Deadline shown in days after the end of each reporting period.

Monthly self-consumption declaration

ES.CNMC.self_consumption_monthly

Declaration of generation and self-consumption to the distribution utility / CNMC.

Recipient: Distributor / CNMC

Cadence: monthly

Format: portal_form

Deadline: 20d after period

Applies when: asset type ∈ SOLAR/WIND/BESS

[RD 244/2019, Art. 14](#)

CECRE real-time telemetry summary

ES.REE.cecre_telemetry

Monthly summary of availability and curtailment events for plants streaming to CECRE.

Recipient: REE (CECRE)

Cadence: monthly

Format: csv

Deadline: 10d after period

Applies when: capacity ≥ 1 MW · asset type ∈ SOLAR/WIND

[P.O. 12.2](#)

Annual production report

ES.CNMC.annual_production

Yearly production & settlement report for plants under regulated remuneration.

Recipient: CNMC

Cadence: annual

Format: pdf

Deadline: 90d after period

Applies when: asset type ∈ SOLAR/WIND/BESS

[RD 413/2014](#)

Safety & inspection cadence

- **OCA (Organismo de Control Autorizado)** · every 60 months · Accredited OCA body · applies when capacity ≥ 0.1 MW

Environmental reporting

- **Guarantees of Origin (GoO)** · monthly to CNMC GoO registry · mandatory

Source documents

- [RD 244/2019 - Self-consumption](#) (verified 2026-04-14)

- [P.O. 12.3 - Technical operation procedures](#) — P.O. 12.3 (verified 2026-04-14)

Portugal (PT)

Regulator: [DGEG](#) · Grid operator: [REN](#)

Pack v2026.Q2 · Currency EUR · Timezone Europe/Lisbon

Portugal follows an ENTSO-E aligned grid code (Portaria 596/2010) with self-consumption governed by DL 15/2022. DGEG is the licensing authority and collector of monthly autoconsumo declarations; REN operates the transmission system and the Guarantees of Origin registry. Technical requirements closely mirror Spain but the reporting cadence and portals differ.

Key terms for Portugal

LVRT — Low Voltage Ride-Through

The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

PCC — Point of Common Coupling

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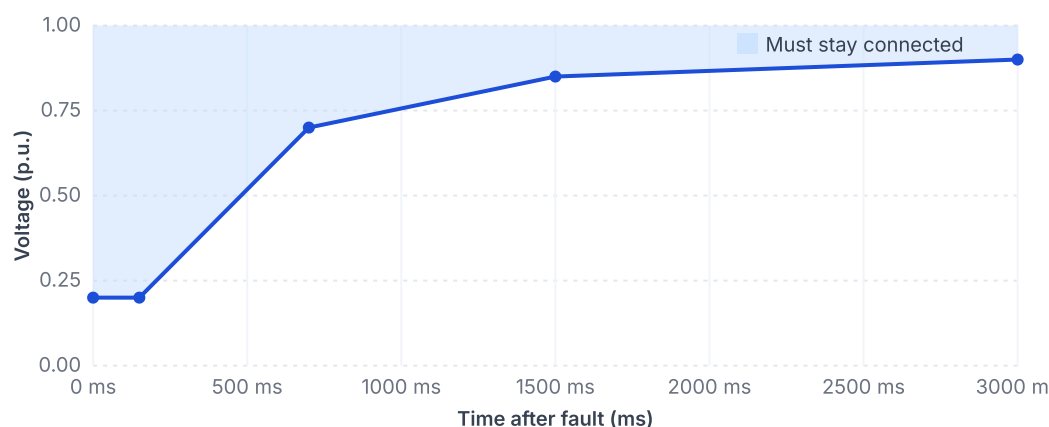
GoO — Guarantees of Origin

EU tradable certificate attesting that 1 MWh of electricity was generated from renewables. Issued via national registries (REN in PT, GSE in IT, CNMC in ES).

Grid connection code

Reference: Portaria 596/2010 / DL 15/2022

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

IEC 62116

RAMP-RATE LIMIT

10% nominal/min

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

10 years

CALIBRATION

every 60 months

Reporting obligations

monthly

Monthly autoconsumo declara...

→ DGEG · 15d

quarterly

Quarterly availability repo...

→ REN · 30d

annual

Annual licence renewal doss...

→ DGEG · 60d

Deadline shown in days after the end of each reporting period.

Monthly autoconsumo declaration

PT.DGEG.autoconsumo_monthly

Self-consumption production declaration to DGEG.

Recipient: DGEG

Cadence: monthly

Format: portal_form

Deadline: 15d after period

Applies when: asset type ∈ SOLAR/WIND/BESS

[DL 15/2022](#)

Quarterly availability report

PT.REN.availability_quarterly

Recipient: REN

Cadence: quarterly

Format: csv

Deadline: 30d after period

Applies when: capacity $\geq$ 1 MW

 [source](#)

Annual licence renewal dossier

PT.DGEG.licence_renewal

Recipient: DGEG

Cadence: annual

Format: pdf

Deadline: 60d after period

Applies when: capacity $\geq$ 1 MW

 [source](#)

Safety & inspection cadence

- **Periodic inspection** · every 60 months · Accredited inspector · applies when capacity $\geq$ 0.1 MW

Environmental reporting

- **Guarantees of Origin (GoO)** · monthly to REN GoO registry · mandatory

Source documents

- [DL 15/2022 - Self-consumption](#) (verified 2026-04-14)
- [Portaria 596/2010 - Grid code](#) (verified 2026-04-14)

Italy has a strict, standards-driven grid code: CEI 0-16 for MV/HV, CEI 0-21 for LV, with an Interface Protection System (SPI) using type-approved relays. ARERA is the overarching regulator; GSE administers incentive schemes (SSP, RID) and collects monthly production data via a dedicated portal. Plants above 1 MW stream telemetry to Terna. Fault-ride-through events must be logged and submitted in the Terna fault logbook.

Key terms for Italy

CEI 0-16

Italian technical standard for MV/HV grid connection. Defines fault ride-through curves, reactive power requirements, and Interface Protection System (SPI).

CEI 0-21

Italian standard for LV grid connection (< 400 V). Simpler envelope than CEI 0-16 — targets residential + small commercial.

Terna

Italian transmission system operator (TSO). Runs the HV grid and publishes grid-code annexes (CEI 0-16).

GSE — Gestore dei Servizi Energetici

Italian state-owned body that administers renewable incentive schemes (SSP, RID) and collects monthly production telemetry.

ARERA — Autorità di Regolazione per Energia Reti e Ambiente

Italian energy, networks, and environment regulator.

SSP — Scambio Sul Posto

Italian "on-site exchange" net-metering scheme administered by GSE for small self-consumption plants.

RID — Ritiro Dedicato

Italian dedicated withdrawal scheme — GSE purchases excess energy from qualifying plants at regulated prices.

DiCo — Dichiarazione di Conformità

Italian declaration of conformity issued by the installer at commissioning. Required for grid connection.

LVRT — Low Voltage Ride-Through

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FRT — Fault Ride-Through

Umbrella term for LVRT + HVRT — staying connected through faults. Prevents simultaneous trips of renewable plants during grid disturbances.

GoO — Guarantees of Origin

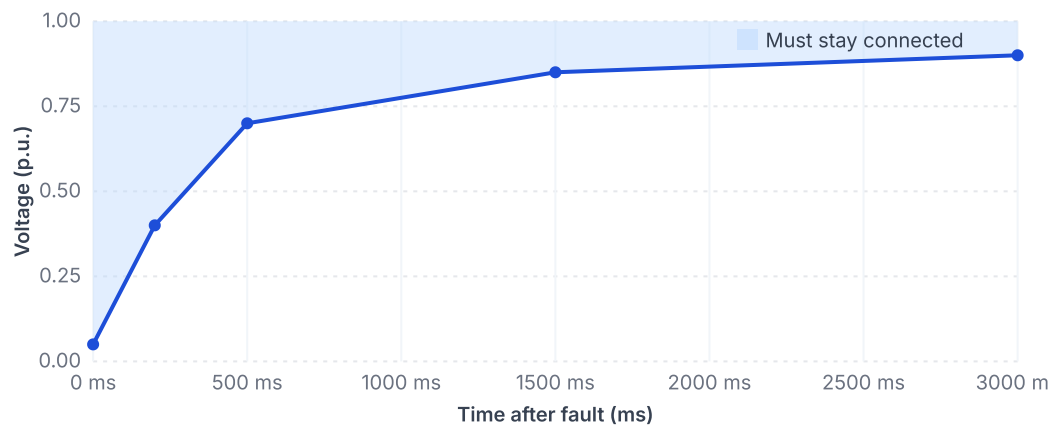
EU tradable certificate attesting that 1 MWh of electricity was generated from renewables. Issued via national registries (REN in PT, GSE in IT, CNMC in ES).

Grid connection code

Reference: CEI 0-16 (MV/HV) / CEI 0-21 (LV)

Interface Protection System (SPI) with certified relays required. Plants >1 MW stream to Terna portal.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.9 to 0.9 at PCC

ANTI-ISLANDING

CEI 0-21 / CEI 0-16

RAMP-RATE LIMIT

20% nominal/min

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

10 years

CALIBRATION

every 60 months

Reporting obligations

event based	Terna fault-ride-through lo... → Terna · 7d
monthly	GSE monthly production tele... → GSE · 15d
annual	ARERA annual quality-of-ser... → ARERA · 90d

Deadline shown in days after the end of each reporting period.

GSE monthly production telemetry

IT.GSE.monthly_production

Monthly production data for plants under SSP (Scambio Sul Posto) or RID (Ritiro Dedicato).

Recipient: GSE

Cadence: monthly

Format: portal_form

Deadline: 15d after period

Applies when: asset type ∈ SOLAR/WIND/BESS

[source](#)

Terna fault-ride-through logbook

IT.TERNA.fault_logbook

Event-based log of LVRT events and Interface Protection activations.

Recipient: Terna

Cadence: event based

Format: csv

Deadline: 7d after period

Applies when: capacity ≥ 1 MW

[CEI 0-16 Annex](#)

ARERA annual quality-of-service report

IT.ARERA.qos_annual

Recipient: ARERA

Cadence: annual

Format: pdf

Deadline: 90d after period

Applies when: capacity ≥ 1 MW

[source](#)

Safety & inspection cadence

- **DiCo (Dichiarazione di Conformità)** · every 0 months · Installer at commissioning
- **ASL safety audit** · every 60 months · ASL (local health authority) · applies when capacity ≥ 0.02 MW

Environmental reporting

- **Guarantees of Origin (GoO)** · monthly to GSE · mandatory

Source documents

- [CEI 0-16 - MV/HV grid connection](#) (verified 2026-04-14)
- [CEI 0-21 - LV grid connection](#) (verified 2026-04-14)
- [GSE portal](#) (verified 2026-04-14)

United Arab Emirates (AE)

Pack v2026.Q2 · Currency AED · Timezone Asia/Dubai

Regulator: [Regulatory & Supervisory Bureau \(RSB\)](#) / [MOEI](#) · Grid operator: [DEWA](#) / [EWEC](#) / [FEWA](#)

The UAE has no unified federal utility — each emirate operates its own distribution company (DEWA in Dubai, EWEC/ADDC in Abu Dhabi, SEWA in Sharjah, FEWA in the Northern Emirates). Dubai's Shams Dubai programme is the dominant distributed-PV framework: net-metered (not feed-in), with IEC 61727 anti-islanding and a Q-U reactive-power curve set in the connection agreement. Reporting is typically done through each utility's own customer portal rather than a central regulator.

Key terms for United Arab Emirates

DEWA — Dubai Electricity & Water Authority

Integrated utility for Dubai; runs distribution + Shams Dubai net-metering.

EWEC — Emirates Water & Electricity Company

Abu Dhabi utility offtaker and single buyer.

Shams Dubai

DEWA's distributed solar programme. Net-metered rooftop/commercial PV with technical requirements defined by DEWA.

RSB — Regulatory & Supervisory Bureau

Dubai regulator for electricity, water, and district cooling.

PCC — Point of Common Coupling

The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

LVRT — Low Voltage Ride-Through

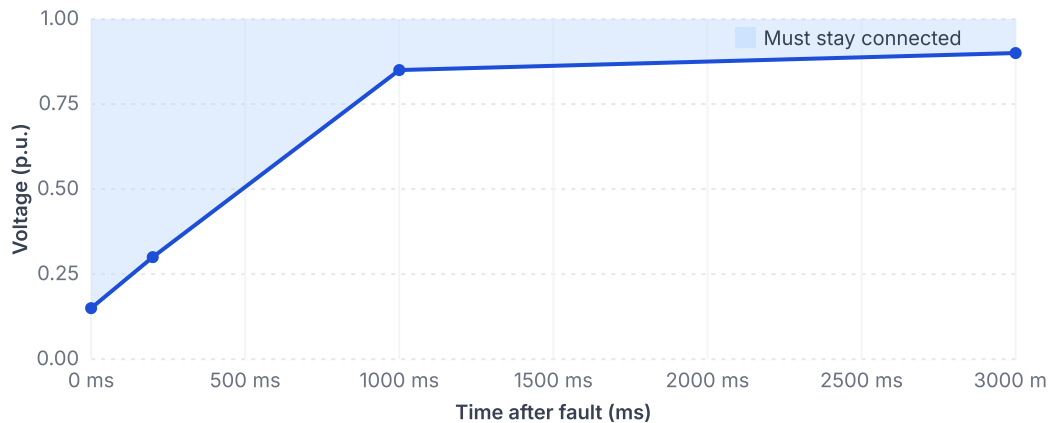
The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

Grid connection code

Reference: DEWA Distribution Code / Shams Dubai Technical Requirements

Net-metering only (no feed-in tariff). Q-U curve mandated per connection agreement.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.9 to 0.9 at PCC

ANTI-ISLANDING

IEC 61727 / IEEE 1547

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

10 years

CALIBRATION

every 60 months

Reporting obligations

monthly

Shams Dubai monthly product...

→ DEWA · 15d

annual

Annual compliance certifica...

→ RSB (Dubai) / EWEC (Abu Dhabi) · 60d

Deadline shown in days after the end of each reporting period.

Shams Dubai monthly production reading

AE.DEWA.shams_monthly

Monthly generation reading submitted via DEWA customer portal for net-metering settlement.

Recipient: DEWA

Cadence: monthly

Format: portal_form

Deadline: 15d after period

Applies when: asset type ∈ SOLAR/BESS

[source](#)

Annual compliance certification

AE.RSB.annual_compliance

Recipient: RSB (Dubai) / EWEC (Abu Dhabi)

Cadence: annual

Format: pdf

Deadline: 60d after period

Applies when: capacity ≥ 1 MW

[source](#)

Safety & inspection cadence

- **Third-party technical audit** · every 12 months · Offtaker-appointed auditor · applies when capacity ≥ 1 MW

Environmental reporting

- **UAE Net Zero 2050 emission-avoidance reporting** · annual to MOEI · voluntary

Source documents

- [DEWA Distribution Code](#) (verified 2026-04-14)
- [Shams Dubai programme](#) (verified 2026-04-14)

Saudi Arabia (SA)

Pack v2026.Q2 · Currency SAR · Timezone Asia/Riyadh

Regulator: [WERA](#) · Grid operator: [SEC / National Grid SA](#)

Saudi Arabia runs a nominal 60 Hz grid. WERA (formerly ECRA) regulates the sector; SEC + National Grid SA operate transmission and distribution. Utility-scale renewables are procured through the NREP auction programme and sold under PPA to the single-buyer SPPC. Distributed rooftop / commercial PV is governed by the SSPV Regulation which uses net billing (offset at wholesale, not retail). Grid-code compliance is independently audited twice per year above 1 MW.

Key terms for Saudi Arabia

WERA — Water and Electricity Regulatory Authority
Saudi Arabian energy regulator (formerly ECRA).

SEC — Saudi Electricity Company
Saudi utility operating transmission and distribution.

SPPC — Saudi Power Procurement Company
Single-buyer offtaker for utility-scale IPPs in Saudi Arabia.

SSPV — Small-Scale Solar PV
Saudi regulatory framework for distributed rooftop/commercial PV with net billing (not net metering).

LVRT — Low Voltage Ride-Through
The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

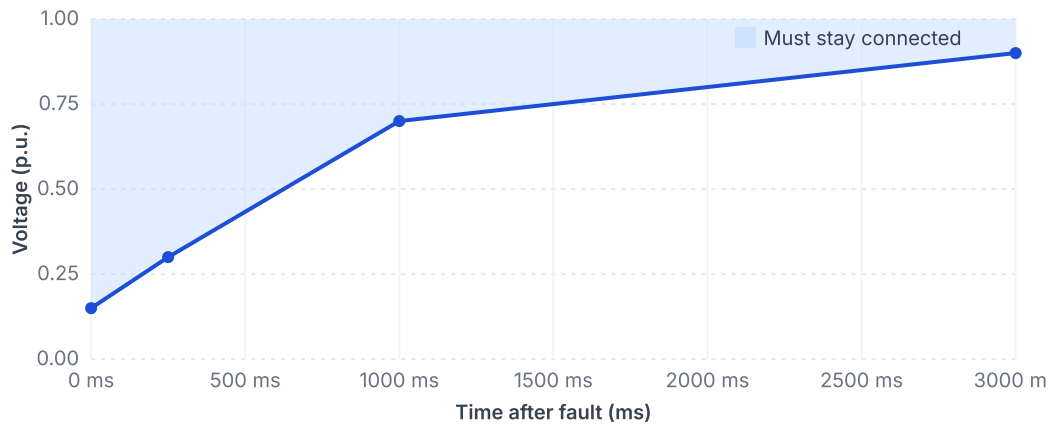
PCC — Point of Common Coupling
The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

Grid connection code

Reference: SEC Grid Code / SSPV Regulation

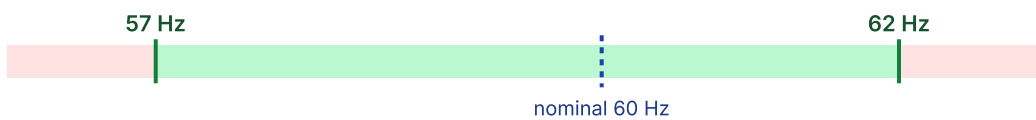
Net billing (not net metering). Strict reactive power follow-me curve mandated for >1 MW.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

IEEE 1547 / IEC 62116

RAMP-RATE LIMIT

10% nominal/min

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

10 years

CALIBRATION

every 60 months

Reporting obligations

monthly	SSPV monthly net-billing re... → SEC · 10d
semi annual	Semi-annual performance rep... → SPPC · 45d Semi-annual technical audit → WERA · 30d

Deadline shown in days after the end of each reporting period.

Semi-annual performance report

SA.SPPC.semi_annual_performance

Performance, availability and curtailment report to the offtaker.

Recipient: SPPC

Cadence: semi annual

Format: pdf

Deadline: 45d after period

Applies when: capacity $\geq$ 1 MW

[source](#)

Semi-annual technical audit

SA.WERA.technical_audit

Grid-code compliance audit: reactive-power curve, protection settings, fault records.

Recipient: WERA

Cadence: semi annual

Format: pdf

Deadline: 30d after period

Applies when: capacity $\geq$ 1 MW

[source](#)

SSPV monthly net-billing reading

SA.SEC.sspv_monthly

Recipient: SEC

Cadence: monthly

Format: portal_form

Deadline: 10d after period

Applies when: capacity $\leq$ 2 MW · asset type $\in$ SOLAR

[source](#)

Safety & inspection cadence

- **Technical audit** · every 6 months · Offtaker-appointed auditor · applies when capacity $\geq$ 1 MW

Environmental reporting

- **Vision 2030 emission-avoidance reporting** · annual to MoEnergy · voluntary

Source documents

- [SEC Grid Code](#) (verified 2026-04-14)
- [SSPV Regulation](#) (verified 2026-04-14)

Kenya (KE)

Pack v2026.Q2 · Currency KES · Timezone Africa/Nairobi

Regulator: [EPRA](#) · Grid operator: [KPLC / KETRACO](#)

Kenya's Energy Act 2019 restructured the sector into separate generation, transmission and distribution functions. EPRA is the sole regulator (tariffs, licensing, grid code); KPLC remains the single buyer under PPAs, with KETRACO owning new transmission. Previously a Feed-in Tariff scheme applied below 10 MW; it is being transitioned into a competitive auction model. Environmental compliance is audited annually by NEMA.

Key terms for Kenya

EPRA — Energy and Petroleum Regulatory Authority
Kenyan energy sector regulator.

KPLC — Kenya Power & Lighting Company
Kenyan single-buyer utility and offtaker for PPAs.

NEMA — National Environment Management Authority
Kenyan environmental regulator — reviews EIAs and annual compliance audits.

LVRT — Low Voltage Ride-Through
The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

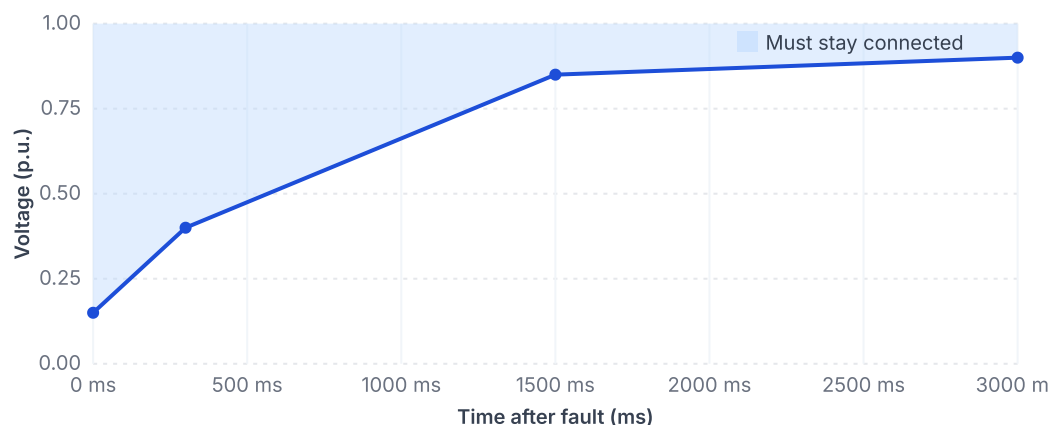
PCC — Point of Common Coupling
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Grid connection code

Reference: EPRA Grid Code 2020 / Energy Act 2019

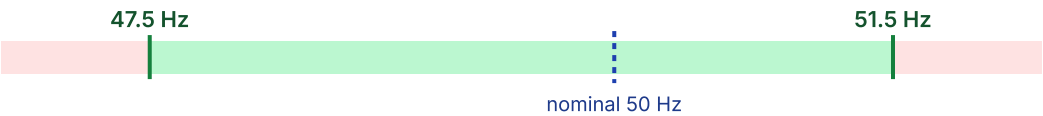
Feed-in Tariff policy applies below 10 MW (transitioning to auction model).

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

IEC 62116

Metering & data retention

METER CLASS
1.0
INTERVAL
15 min
RETENTION
5 years
CALIBRATION
every 36 months

Reporting obligations

monthly	KPLC PPA monthly settlement... → KPLC - 10d
quarterly	EPRA quarterly generation p... → EPRA - 30d
annual	NEMA annual environmental c... → NEMA - 90d

Deadline shown in days after the end of each reporting period.

EPRA quarterly generation performance report Quarterly report on generation, availability, and outage events. Recipient: EPRA Cadence: quarterly Format: pdf Deadline: 30d after period Applies when: asset type ∈ SOLAR/WIND/BESS source	KE.EPRA.quarterly_performance
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KPLC PPA monthly settlement data

KE.KPLC.ppa_monthly

Recipient: KPLC

Cadence: monthly

Format: csv

Deadline: 10d after period

Applies when: capacity $\geq$ 1 MW

[source](#)

NEMA annual environmental compliance audit

KE.NEMA.annual_environmental

Recipient: NEMA

Cadence: annual

Format: pdf

Deadline: 90d after period

Applies when: capacity $\geq$ 0.5 MW

[source](#)

Safety & inspection cadence

- **EPRA electrical inspection** · every 36 months · EPRA · applies when capacity $\geq$ 0.1 MW

Environmental reporting

- **NEMA environmental monitoring** · annual to NEMA · mandatory

Source documents

- [Energy Act 2019](#) (verified 2026-04-14)
- [EPRA Grid Code 2020](#) (verified 2026-04-14)

Nigeria (NG)

Regulator: [NERC](#) · Grid operator: [TCN](#)

Pack v2026.Q2 · Currency NGN · Timezone Africa/Lagos

Nigeria's NESI (Nigerian Electricity Supply Industry) is unbundled into NERC (regulator), TCN (transmission), eleven distribution companies, and NBET as the single-buyer bulk trader for IPPs. The Mini-Grid Regulation 2016 governs off-grid/rural systems below 1 MW — a distinct track from large-grid PPAs. All new plants require a NEMSA pre-energisation safety inspection before they are allowed to synchronise.

Key terms for Nigeria

NERC — Nigerian Electricity Regulatory Commission
Federal regulator for the Nigerian electricity industry.

NBET — Nigerian Bulk Electricity Trading
Single-buyer bulk trader that signs PPAs with IPPs in Nigeria.

NEMSA — Nigerian Electricity Management Services Agency
Nigerian electrical safety agency — mandates pre-energisation inspection.

LVRT — Low Voltage Ride-Through
The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

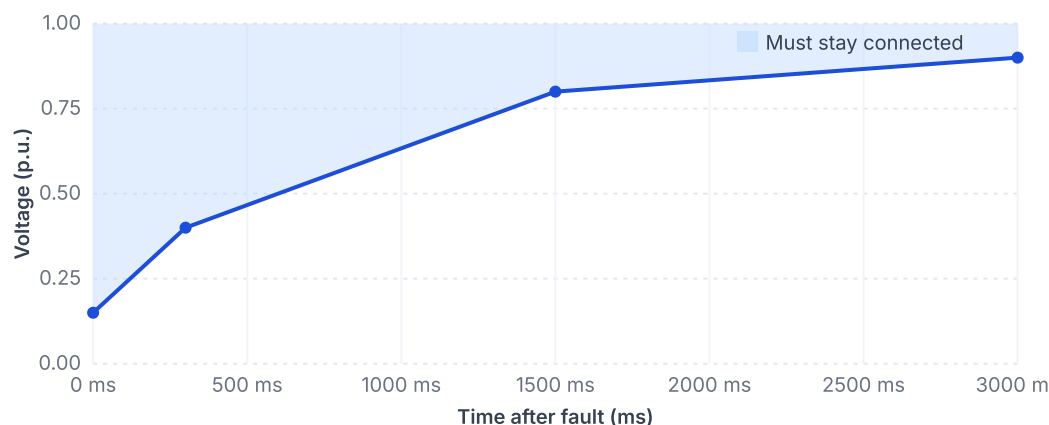
PCC — Point of Common Coupling
The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

Grid connection code

Reference: NERC Grid Code / Distribution Code / Mini-Grid Regulation 2016

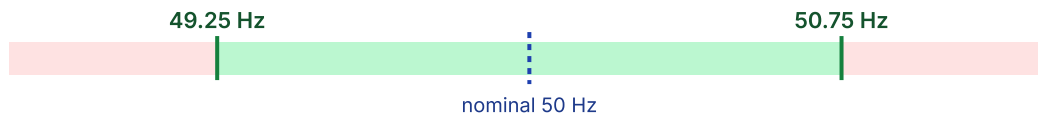
NESI Metering Code (MAP) applies; NEMSA inspection mandatory before energisation.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

IEC 62116

Metering & data retention

METER CLASS

1.0

INTERVAL

15 min

RETENTION

5 years

CALIBRATION

every 36 months

Reporting obligations

event based	NEMSA pre-energisation insp... → NEMSA : 0d
monthly	NERC monthly generation & a... → NERC : 15d
quarterly	NBET quarterly PPA report → NBET : 30d

Deadline shown in days after the end of each reporting period.

NERC monthly generation & availability report

NG.NERC.monthly_generation

Monthly generation, availability and outage report.

Recipient: NERC

Cadence: monthly

Format: pdf

Deadline: 15d after period

Applies when: asset type ∈ SOLAR/WIND/BESS

[source](#)

NBET quarterly PPA report

NG.NBET.quarterly_ppa

Recipient: NBET

Cadence: quarterly

Format: pdf

Deadline: 30d after period

Applies when: capacity $\geq$ 1 MW

[source](#)

NEMSA pre-energisation inspection record

NG.NEMSA.pre_energisation

Recipient: NEMSA

Cadence: event based

Format: pdf

Deadline: 0d after period

[source](#)

Safety & inspection cadence

- **NEMSA periodic inspection** · every 24 months · NEMSA · applies when capacity $\geq$ 0.1 MW

Environmental reporting

- **Environmental impact assessment (EIA) monitoring** · annual to Federal Ministry of Environment · mandatory

Source documents

- [NERC Grid Code](#) (verified 2026-04-14)
- [Mini-Grid Regulation 2016](#) (verified 2026-04-14)

South Africa (ZA)

Pack v2026.Q2 · Currency ZAR · Timezone Africa/Johannesburg

Regulator: [NERSA](#) · Grid operator: [Eskom](#)

South Africa splits distributed generation into two regimes: small-scale embedded generation (SSEG) up to 100 kVA, governed by NRS 097-2-1 / -3 and connected via Eskom or the local municipality, and Renewable Power Plants (RPP), categorised A1 through C by capacity with escalating SCADA, protection and reporting requirements from Category B upward. NERSA licenses generators above 100 MW; utility-scale IPPs participate in REIPPPP / RMIPPPP and submit monthly dispatch data to DMRE. A Certificate of Compliance (COC) is required at energisation and after any modification.

Key terms for South Africa

NERSA — National Energy Regulator of South Africa
South African electricity, gas, and piped-fuel regulator.

SSEG — Small-Scale Embedded Generation
South African category for distributed generation ≤ 1 MW connected at LV/MV. Governed by NRS 097-2-1 and NRS 097-2-3.

NRS 097
South African standard series for grid connection of embedded generation. -2-1 covers utility interface, -2-3 covers simplified registration for ≤ 100 kVA.

RPP — Renewable Power Plant
South African grid-code category covering larger renewable plants (> 1 MW). Sub-categorised A1/A2/A3/B/C by capacity with escalating requirements.

REIPPPP — Renewable Energy Independent Power Producer Procurement Programme
South African competitive procurement programme for utility-scale renewables, overseen by DMRE.

COC — Certificate of Compliance
South African post-installation certificate issued by an accredited electrician. Must be revalidated after modifications.

LVRT — Low Voltage Ride-Through
The ability of a generator to stay connected to the grid during a voltage dip without tripping. Defined as a curve of voltage (per-unit) vs. time after fault.

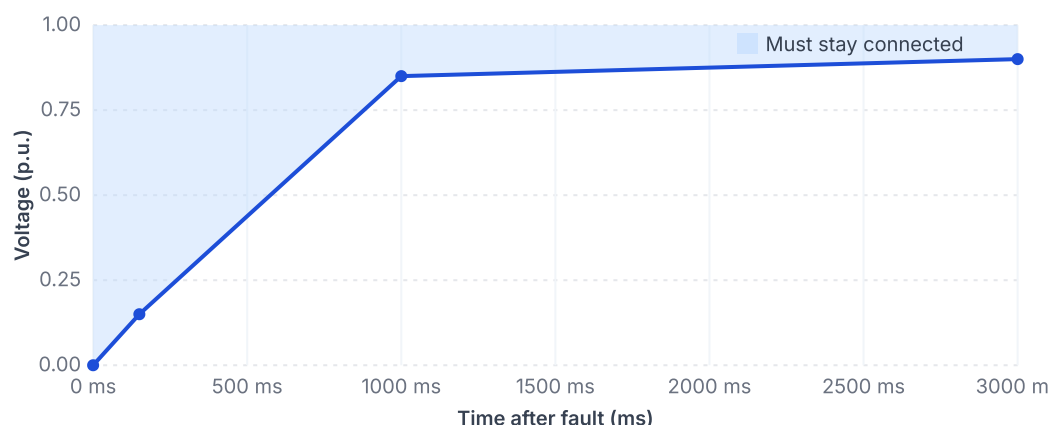
PCC — Point of Common Coupling
The electrical boundary where the plant connects to the public grid. Grid code requirements (voltage, reactive power, harmonics) are measured here.

Grid connection code

Reference: RPP Grid Code / NRS 097-2-1 / NRS 097-2-3

Category B+ plants must provide a SCADA link to the system operator. COC required; revalidated after any modification.

Low-Voltage Ride-Through (LVRT) envelope



Voltage dips on or above the blue line: the plant must ride through without tripping. Dips below the line: disconnection permitted.

Frequency operating window



Green = must stay connected. Red = disconnection permitted by protection settings.

REACTIVE POWER

Power factor -0.95 to 0.95 at PCC

ANTI-ISLANDING

NRS 097-2-1 / IEC 62116

RAMP-RATE LIMIT

10% nominal/min

Metering & data retention

METER CLASS

0.5S

INTERVAL

15 min

RETENTION

5 years

CALIBRATION

every 60 months

Reporting obligations

monthly	DMRE monthly dispatch data ... → DMRE · 10d
quarterly	NERSA quarterly compliance ... → NERSA · 30d
annual	NRS 097 SSEG compliance che... → Eskom / municipality · 60d

Deadline shown in days after the end of each reporting period.

NERSA quarterly compliance report

ZA.NERSA.quarterly_compliance

Quarterly compliance report for licensed generators.

Recipient: NERSA

Cadence: quarterly

Format: pdf

Deadline: 30d after period

Applies when: capacity $\geq$ 1 MW

[source](#)

DMRE monthly dispatch data (RMIPPPP / REIPPPP)

ZA.DMRE.monthly_dispatch

Recipient: DMRE

Cadence: monthly

Format: csv

Deadline: 10d after period

Applies when: capacity $\geq$ 10 MW

[source](#)

NRS 097 SSEG compliance checklist

ZA.NRS.097_checklist

Checklist confirming compliance with NRS 097-2-1 / -3 for small-scale embedded generation.

Recipient: Eskom / municipality

Cadence: annual

Format: pdf

Deadline: 60d after period

Applies when: capacity $\leq$ 1 MW

[source](#)

Safety & inspection cadence

- **COC (Certificate of Compliance)** · every 0 months · Accredited electrician (post-modification)

Environmental reporting

- **REIPPPP carbon offset reporting** · annual to DMRE · voluntary

Source documents

- [Grid Connection Code for Renewable Power Plants](#) (verified 2026-04-14)

- [NRS 097-2-1 / NRS 097-2-3](#) (verified 2026-04-14)

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